

35—BAR SCALES (continued)

BAR SCALE CALCULATIONS — MILES (1 mile = 63,360 inches)							
FRACTIONAL SCALE	SCALE TO MAP REPRESENTATION		TO FIND MILES PER INCH (x in ratio) Use ratio below or $\frac{\text{SCALE}}{63\,360}$	MILES PER INCH	TOTAL MILES ON SCALE	TO FIND TOTAL SCALE LENGTH IN INCHES (y in ratio) Use ratio below or $\frac{\text{Miles on scale}}{\text{Miles per inch}}$	TOTAL SCALE LENGTH (INCHES)
	Scale Unit :	represents Map Unit					
1:12 000	1 inch	12 000 in	$\frac{63\,360}{1} = \frac{12\,000}{x}$	0.1893939	1.5	$\frac{0.1893939}{1} = \frac{1.5}{y}$	7.920
1:20 000	1 inch	20 000 in	$\frac{63\,360}{1} = \frac{20\,000}{x}$	0.3156565	2	$\frac{0.3156565}{1} = \frac{2}{y}$	6.336
1:24 000	1 inch	24 000 in	$\frac{63\,360}{1} = \frac{24\,000}{x}$	0.3787878	2	$\frac{0.3787878}{1} = \frac{2}{y}$	5.280
1:25 000	1 inch	25 000 in	$\frac{63\,360}{1} = \frac{25\,000}{x}$	0.3945707	2	$\frac{0.3945707}{1} = \frac{2}{y}$	5.068
1:31 250	1 inch	31 250 in	$\frac{63\,360}{1} = \frac{31\,250}{x}$	0.4932133	2	$\frac{0.4932133}{1} = \frac{2}{y}$	4.055
1:31 680	1 inch	31 680 in	$\frac{63\,360}{1} = \frac{31\,680}{x}$	0.500	2	$\frac{0.500}{1} = \frac{2}{y}$	4.000
1:48 000	1 inch	48 000 in	$\frac{63\,360}{1} = \frac{48\,000}{x}$	0.7575757	5	$\frac{0.7575757}{1} = \frac{5}{y}$	6.600
1:50 000	1 inch	50 000 in	$\frac{63\,360}{1} = \frac{50\,000}{x}$	0.7891414	5	$\frac{0.7891414}{1} = \frac{5}{y}$	6.336
1:62 500	1 inch	62 500 in	$\frac{63\,360}{1} = \frac{62\,500}{x}$	0.9864267	6	$\frac{0.9864267}{1} = \frac{6}{y}$	6.082
1:63 360	1 inch	63 360 in	$\frac{63\,360}{1} = \frac{63\,360}{x}$	1.000	6	$\frac{1.000}{1} = \frac{6}{y}$	6.000
1:72 000	1 inch	72 000 in	$\frac{63\,360}{1} = \frac{72\,000}{x}$	1.1363636	6	$\frac{1.1363636}{1} = \frac{6}{y}$	5.280
1:75 000	1 inch	75 000 in	$\frac{63\,360}{1} = \frac{75\,000}{x}$	1.1837121	6	$\frac{1.1837121}{1} = \frac{6}{y}$	5.068
1:96 000	1 inch	96 000 in	$\frac{63\,360}{1} = \frac{96\,000}{x}$	1.5151515	7	$\frac{1.5151515}{1} = \frac{7}{y}$	4.620
1:100 000	1 inch	100 000 in	$\frac{63\,360}{1} = \frac{100\,000}{x}$	1.5782828	9	$\frac{1.5782828}{1} = \frac{9}{y}$	5.702
1:125 000	1 inch	125 000 in	$\frac{63\,360}{1} = \frac{125\,000}{x}$	1.9728535	12	$\frac{1.9728535}{1} = \frac{12}{y}$	6.082
1:150 000	1 inch	150 000 in	$\frac{63\,360}{1} = \frac{150\,000}{x}$	2.3674242	12	$\frac{2.3674242}{1} = \frac{12}{y}$	5.068
To find miles per inch on 1: 12 000 map . . . 63,360 inches = 1 mile Show in ratio as ... $\frac{63\,360}{1} \text{ inches}$ 1 miles Let SCALE (12 000) be in inches Fractional scale says 1 inch represents 12,000 in Let x be miles that 1 inch represents on map Show in ratio as ... $\frac{12\,000}{x} \text{ inches}$ $x \text{ miles}$ Solution . . . $\frac{63\,360}{1} = \frac{12\,000}{x}$ $\frac{63\,360 \cdot x}{63\,360} = \frac{12\,000}{63\,360}$ $x = \frac{12\,000}{63\,360} \text{ (SCALE)}$ $x = 0.1893939$							